GITAM Deemed to be University

Computer Science and Engineering | Course Code: CS101

Object-Oriented Programming & Principles - Laboratory Manual & Practical

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Syllabus Focus: CS101 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Paradigms & Object-Oriented Fundamentals

" Syllabus Scope: Procedural vs Object-Oriented paradigms, Data encapsulation, Access specifiers, Constru

Unit 2: Inheritance & Dynamic Binding

" Syllabus Scope: Single, multiple, and hierarchical inheritance, Diamond problem, Virtual base classes,

CS101: Object-Oriented Programming & Principles - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Memory Architecture & RAII

" Stack vs Heap memory allocation, RAII idiom, Smart pointers unique_ptr, shared_ptr, weak_ptr, Generic templa

Unit 4: Exception Hierarchies & Stream I/O

" Try-catch-throw mechanisms, Custom exception classes, File streams, Serialization, Binary and formatted disk

Unit 5: Modern Language Idioms & Design

" Lambda expressions, Move semantics, Rvalue references, Operator overloading, Thread concurrency primitives

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Implementation of Classes, Static Members, and Overloaded Constructors in C++

Lab 2: Polymorphic Hierarchy Demonstration with Virtual Functions and Abstract Base Classes

Lab 3: Custom Generic Stack and Queue Template Classes with Exception Handling

Lab 4: Binary File Persistence and Object Serialization using C++ File Streams

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Bjarne Stroustrup - The C++ Programming Language Addison-Wesley 4th Ed

* Robert Lafore - Object-Oriented Programming in C++ Sams Publishing

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